# Colloquium Class

# What will we do in this class?

- Attend weekly colloquium talks
- Talks will be delivered by scientists from:
  - A variety of physics sub-fields
  - Both inside and outside IIT
  - Both inside and outside physics
- Engage with the speaker and audience by asking questions
- Eat cookies and drink coffee beforehand (usually…)!
- Perform some writing assignments related to topics we have heard about in colloquium

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- <u>Observing</u> effective (and ineffective) scientific communication
- <u>Practicing</u> your own scientific writing/communication skills
- Generating regular social engagement within the department
    - Related: Maintaining sufficient attendance to ensure a vibrant weekly colloquium can be offered in the department

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# Scientific Communication

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- Colloquium speakers are sharing physics with you
  - So you think about colloquia being about the PHYSICS more than anything else
- Some speakers will be more effective sharers than others
  - WHY???
  - This impacts the amount of physics that actually sticks with you beyond the talk!
  - So clearly the methods of communication
- In this class, let's try to take in both the physics <u>and</u> the scientific communication strategies of speakers

# Scientific Communication Final

- Get a notebook specifically for this class — or organize notes for each colloquium on your computer.
- During colloquia, take note of:
  - What was the speaker talking about? What was their topic?
  - What communication techniques/styles did speakers use were effective? Why was it effective?
  - What did speakers do that was <u>not</u> effective, or confusing?
- During your 'final exam' period, we will all perform an exercise together to share our impressions about effective and ineffective scientific communication strategies.
  - The goal here would be for you to identify strategies that <u>YOU</u> can use in YOUR scientific talks and presentations in the future!

# Scientific writing:
# Yes, it's a thing you all need to learn.

Snakes on a Spaceship—An Overview of Python in Heliophysics

A. G. Burrell, A. Halford, J. Klenzing, R. A. Stoneback, S. K. Morley ... See all authors ›

miR miR on the wall, who's the most malignant medulloblastoma miR of them all?

Review article

Wang X, et al. Neuro Oncol. 2018.

Understanding the Dynamics of Emerging and Re-Emerging Infectious Diseases Using Mathematical Models, 2012: 157-177 ISBN: 978-81-7895-549-0 Editors: Steady Mushayabasa and Claver P. Bhunu

7. A mathematical model of Bieber Fever: The most infectious disease of our time?

Valerie Tweedle[1] and Robert J. Smith?[2]

# Scientific Writing

- **<u>I cannot stress enough how central scientific writing is to being a successful physicist.</u>**
- How many of you have taken a scientific writing course?
- If you can't write well:
  - You can't write a <u>research/personal</u> statement well, which makes getting a grad, postdoc, or scientist position difficult.
  - You can't write a <u>scientific publication</u> well, which will make understanding and publishing your groundbreaking new physics discovery difficult.
  - You can't write a <u>funding proposal</u> well, which makes getting research funding difficult.
- As a physics Prof, writing is one of the things I do the most!

# Scientific Writing Resources

- This is a 1-credit course, so there's only so much time…
- Resources:
  - https://www.nature.com/articles/d41586-018-02404-4
  - https://www.elsevier.com/connect/11-steps-to-structuring-a-science-paper-editors-will-take-seriously
  - https://writing.wisc.edu/handbook/process/reverseoutlines/
  - http://www.physik.uni-regensburg.de/forschung/fabian/pages/mainframes/lecturenotes/lecturenotes_files/writing_physics_papers.pdf
  - https://www.urbanphysics2013.org/prompts-for-outlining-a-physics-research-paper-properly
- Let's go over a few highlights:
  - Organization!
  - Creativity!
  - Cut the filler!
  - Broaden the audience!

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  - https://www.elsevier.com/connect/11-steps-to-structuring-a-science-paper-editors-will-take-seriously
  - https://writing.wisc.edu/handbook/process/reverseoutlines/
  - http://www.physik.uni-regensburg.de/forschung/fabian/pages/mainframes/lecturenotes/lecturenotes_files/writing_physics_papers.pdf
  - https://www.urbanphysics2013.org/prompts-for-outlining-a-physics-research-paper-properly
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# Scientific Writing Assignment, All

- Reverse outlining: a strategy for organized writing
- <u>We will, as a class, reverse-outline a scientific paper</u>
    - We will pick a seminal paper related the Nobel Prize in Physics (awarded in October)
    - We will all read this paper
    - You each will then perform a reverse outlining of that paper
    - Working together with classmates is encouraged!!!
    - After the Nobel Colloquium, we will work through our reverse outlines together
- In case you don't understand the process, I will provide a tutorial video on the BlackBoard for you ahead of time!
- If you know can effectively reverse-outline, you can effectively turn one of <u>your own</u> outlines into a well-organized paper!
- Turn in your outline on the BlackBoard!

# Scientific Writing Assignments, Grads

- Write two ArXiv article reviews: https://arxiv.org/
  - See here for details: http://phys.iit.edu/~segre/phys585-685/
  - Identify a RECENT (in last 2 years!) article (not proceedings!) manuscript that was related to one of the speaker's research.
  - Easy way to pick a 'good article': use an article that was cited by that speaker in their talk!
  - Get it OK'ed by me
  - Write a 1-2 page summary of the paper, including a description of the paper's strategy, results and takeaways, recommendations for improvement, and a final judgement whether to ACCEPT, REVISE, or REJECT the paper for publication!
- I have an example of a 'good review' on the BlackBoard for you
- Use that outlining strategy that you will have practiced to generate an organized review write-up!

# Scientific Writing Assignments, Undergrads

- Write <u>three</u> colloquium reviews
  - See here for details: http://phys.iit.edu/~segre/phys485/
  - I will decide which weeks will require a colloquium review. I'll usually choose a speaker that seemed to me to be pretty engaging and accessible.
  - Write a 1-2 page summary of the colloquium, including a description of the broad topic and specific results discussed by the speaker, items of the talk that you found most interesting, and your assessment of the speaker's communication strategies.
- I have an example of a 'good review' on BlackBoard for you
- Use that outlining strategy that you will have practiced to generate an organized review write-up!

Questions?